
Same Top Fraction, Different Workload: Auditing Eligibility and Capacity in Clinical Risk Evaluation

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Abstract

1 Clinical risk scores turn patient rankings into finite worklists for specialist review,
2 monitoring, or prevention. Yet cross-cohort evaluations often compare the same
3 top fraction across candidate sets of different sizes, changing both who can be
4 ranked and how many people receive service. We introduce a decision-aware four-
5 cell audit that keeps the full-population event denominator fixed and crosses two
6 eligibility rules with two absolute budgets, separating broader reach from addi-
7 tional capacity. We apply it to frozen birth-record score vectors for preterm birth,
8 neonatal intensive care admission, and low birth weight in approximately 3.48 mil-
9 lion CDC Natality 2023 records. At the nominal top 10% operating point, moving
10 from early-entry eligibility to a retrospective all-record upper bound reports a 39–
11 42% relative increase in captured events while expanding the worklist by one third.
12 Symmetric accounting allocates about three fifths of the joint gain to added capac-
13 ity and two fifths to broader eligibility; at a fixed budget, the eligibility gain is
14 less than half the cohort-relative estimate. In 22% of controlled conditions, the
15 cohort-relative comparison improves while fixed-budget capture declines. The au-
16 dit therefore distinguishes population reach from resource expansion and makes
17 the policy represented by a top-fraction metric explicit.

18 1 Introduction

19 Clinical risk models increasingly determine which records receive specialist review, monitoring,
20 outreach, or preventive services [Annarumma et al., 2019, Baltruschat et al., 2021, Jung et al., 2021,
21 Singh et al., 2023]. These workflows are capacity constrained: a model produces a ranking, but an or-
22 ganization acts on a finite number of cases. Top- q precision, recall, or event capture therefore appears
23 to connect predictive performance directly to an operational decision. Prenatal case-management
24 studies, for example, evaluate the highest-risk 5–15% because program capacity covers a fraction of
25 eligible pregnancies [Pan et al., 2017].

26 The concrete scale of this policy difference is visible in CDC Natality 2023 [National Center for
27 Health Statistics, 2026]. An early-entry candidate set of approximately 2.60 million records yields a
28 top-10% worklist of about 260,000, whereas an all-record set of approximately 3.48 million yields
29 about 348,000 selections at the same nominal fraction. The comparison therefore changes both
30 the candidate gate and the absolute workload. Clinical decision frameworks distinguish predictive
31 ranking from action under finite resources [Jung et al., 2021, Singh et al., 2023]: a fixed-capacity
32 service and a service that expands with candidate volume implement different policies and require
33 different contrasts.

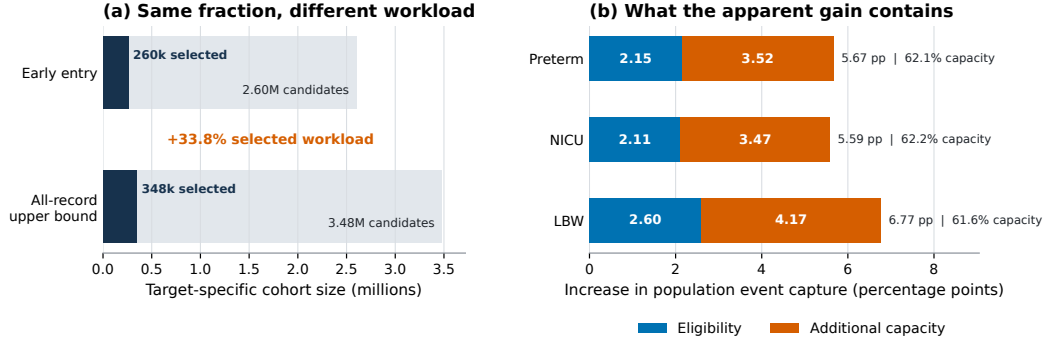


Figure 1: The same nominal top 10% encodes a different workload. Panel (a) summarizes the target-specific CDC 2023 cohorts: moving from the early-entry gate to the all-record upper bound increases the selected count from about 260,000 to 348,000. Panel (b) allocates the associated increase in population event capture at $q = 10\%$. Values inside bars are percentage-point allocations; capacity contributes about 62% of each total gain.

Prenatal risk triage makes this ambiguity visible. Pregnancies enter care at different times, and adverse outcomes are concentrated among records with no documented prenatal care or unknown start timing. An early-entry score ranks events inside its candidate gate; events outside the gate contribute zero reachable cases regardless of discrimination. National vital statistics retain both the eligible and excluded outcomes, making it possible to distinguish retrieval within the gate from reach across the complete target population.

We audit this distinction on CDC Natality preterm birth, neonatal intensive care unit (NICU) admission, and low birth weight. Figure 1 previews the result. At the nominal top 10% operating point, all-record upper-bound evaluation selects roughly 88,000 additional records per outcome. Population event capture rises substantially, but about 62% of the joint increase is allocated to the larger workload. A controlled mechanism grid then identifies conditions in which the resource change is large enough to reverse the fixed-budget conclusion.

Our contributions are:

1. **An auditable decision estimand.** Population event capture keeps every target-population event in the denominator, while a complete eligibility-by-budget design exposes the two fixed-budget contrasts and their interaction.
2. **A national-scale measurement study.** Frozen 2022 scoring pipelines applied to CDC 2023 show that the same top 10% expands workload by 33.8% and that capacity receives 61.6–62.2% of the symmetric allocation across three outcomes.
3. **A mechanism map and reporting profile.** An 81-condition stress test locates sign-reversal regimes, and the resulting checklist makes the candidate gate, absolute workload, and event denominator explicit.

Code for the four-cell audit and synthetic experiments, together with the aggregate artifacts used to generate the reported figures and tables, will be released upon publication.

2 Related Work

Risk ranking under limited resources. Prenatal case-management, imaging worklists, and clinical workflow studies connect risk scores to finite review or treatment capacity [Pan et al., 2017, Annarumma et al., 2019, Baltruschat et al., 2021, Jung et al., 2021, Singh et al., 2023]. They motivate operating points such as top 5%, top 10%, or a fixed number of patients and show why use-aligned metrics matter. Calibrated subset selection similarly constructs shortlists with guarantees inside a supplied candidate pool [Wang et al., 2022]. These approaches make selection within a declared pool more reliable, while leaving cross-pool comparisons dependent on how the shortlist size changes with the pool.

Selective access to clinical data and services. Clinical records reflect who enters care, is tested, and receives a documented label. Informed presence and selection bias can alter both apparent risk and the population represented by an EHR cohort [Haneuse and Daniels, 2016, Goldstein et al., 2016, Sisk et al., 2021]; disparate censorship describes related variation in testing and label availability [Chang et al., 2022]. Health-service coverage frameworks separate service availability from realized coverage [Tanahashi, 1978]. This literature establishes that the candidate gate is informative, but a top-fraction comparison adds a second moving quantity because the number served scales with the observed cohort.

Evaluation validity and decomposition. Clinical reporting guidance requires explicit target populations, intended use, discrimination, and calibration [Collins et al., 2024]. Symmetric allocation across factor orderings is established decomposition machinery [Kitagawa, 1955, Shapley, 1953, Shorrocks, 2013]. Taken together, these conventions still leave the operational meaning of a metric implicit when candidate membership and absolute capacity change together. The case is common and concrete: a percentage-based budget is held numerically constant across cohorts, its absolute size moves, and the resulting gain is reported as if it represented one intervention.

3 Methodology

The audit fixes the target population, score vector, and event denominator, then crosses two candidate-set rules with two absolute budgets. Figure 2 shows the resulting four cells and the two paths whose average yields the symmetric allocation.

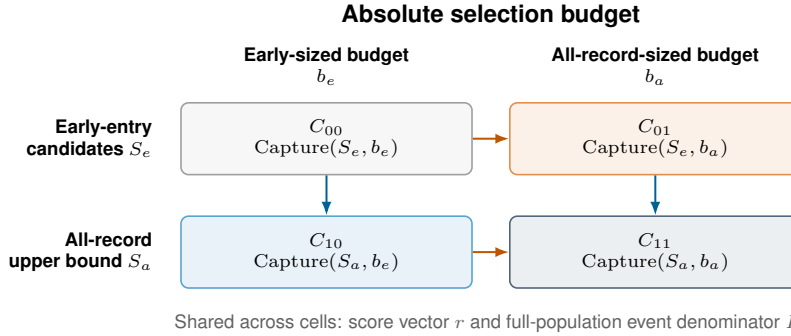


Figure 2: Eligibility–capacity audit. Rows vary the candidate set and columns vary the absolute selection budget while the score vector and full-population event denominator remain fixed. Blue arrows change candidate eligibility; orange arrows change capacity. Averaging the two paths from C_{00} to C_{11} gives the symmetric Shapley allocation, whereas $C_{11} - C_{00}$ is the usual same- q comparison and changes both factors.

3.1 Target population and scoring study

We use CDC Natality public-use live-birth records and official field definitions [National Center for Health Statistics, 2026, 2017]. CDC 2022 supplies model development with an 80/20 train/validation split; the frozen pipelines generate the shared score vectors analyzed in CDC 2023. Model fitting and hyperparameter selection use CDC 2022 exclusively; CDC 2023 is the temporal cohort in which the audit hypothesis was developed and evaluated. The target population contains U.S.-resident singleton live births with a known outcome: preterm birth below 37 completed obstetric-estimate weeks ($N = 3,480,382$, 303,001 events), NICU admission ($N = 3,476,584$, 303,285 events), and birth weight below 2,500 g ($N = 3,480,316$, 243,776 events).

Prenatal-care month defines four mutually exclusive groups: early entry in months 1–3, later documented entry in months 4–10, no prenatal care, and unknown start timing. The early candidate set contains months 1–3. The documented-start set contains months 1–10. The all-record set adds no-care and unknown-timing records and therefore bounds candidate and event availability under retrospective complete scoring. The care-month rule operationalizes documented entry into prenatal

care rather than feature-level EHR availability; the birth-certificate model serves as a shared retrospective ranking instrument.

The baseline-oriented score uses nine maternal birth-certificate variables: race, height, pre-pregnancy BMI and weight, previous live births now living, previous live births now dead, other pregnancy outcomes, pre-pregnancy diabetes, and pre-pregnancy hypertension. Outcome, delivery, and neonatal fields are excluded. Numeric variables use development-set median imputation and standardization; race uses most-frequent imputation and one-hot encoding. One LightGBM model per outcome uses 350 trees, learning rate 0.05, 31 leaves, minimum child size 100, and seed 2026 [Ke et al., 2017]. The audit compares candidate and budget policies on a shared score vector while holding the model family constant.

3.2 Decision metrics

Let $\mathcal{P}$ contain N records and $E = \sum_{i \in \mathcal{P}} Y_i$ events. For candidate set S , event count E_S , and integer budget b , deterministic tie breaking defines the b highest scores $\text{Top}(S, b)$. Their true-positive count is

$$\text{TP}(S, b) = \sum_{i \in \text{Top}(S, b)} Y_i. \quad (1)$$

We report both within-gate performance and population reach:

$$\text{Precision}(S, b) = \frac{\text{TP}(S, b)}{b}, \quad \text{Recall}_{\text{eligible}}(S, b) = \frac{\text{TP}(S, b)}{E_S}, \quad (2)$$

$$\text{Capture}(S, b) = \frac{\text{TP}(S, b)}{E}, \quad \text{Ceiling}(S) = \frac{E_S}{E}. \quad (3)$$

Population capture factors into retrieval and reach,

$$\text{Capture}(S, b) = \text{Recall}_{\text{eligible}}(S, b) \text{ Ceiling}(S). \quad (4)$$

Each result therefore reports target-population N and E , eligible n and E_S , exact selected n , precision, eligible recall, population capture, and the event ceiling. ROC-AUC and average precision (AP) provide a ranking-signal check; the four-cell contrasts carry the primary decision claim.

3.3 Four-cell eligibility–capacity audit

Let $S_e \subseteq S_a$ be early and all-record candidate sets. A nominal fraction q induces $b_e = \text{round}(q|S_e|)$ and $b_a = \text{round}(q|S_a|)$. With $b_a \leq |S_e|$, the complete design is

$$C_{uv} = \text{Capture}(S_u, b_v), \quad u, v \in \{0, 1\}, \quad (5)$$

where 0 denotes early and 1 denotes all-record eligibility or capacity. The usual cohort-relative contrast is $\Delta = C_{11} - C_{00}$. Its interaction is $I = C_{11} - C_{10} - C_{01} + C_{00}$. We retain both fixed-budget eligibility contrasts and summarize the joint change with the standard symmetric Shapley allocation [Shapley, 1953]:

$$\begin{aligned} \phi_E &= \frac{1}{2} [(C_{10} - C_{00}) + (C_{11} - C_{01})], \\ \phi_B &= \frac{1}{2} [(C_{01} - C_{00}) + (C_{11} - C_{10})]. \end{aligned} \quad (6)$$

The exact identity $\Delta = \phi_E + \phi_B$ holds, and $s_B = \phi_B / \Delta$ gives the symmetric capacity share for positive Δ . For a frozen deterministic ranking, increasing b adds nested positions, so the capacity component is non-negative; expanding S at fixed b can have either sign. A uniformly random ranking supplies the mechanical reference $E[\text{Capture}(S, b)] = (b/|S|)(E_S/E)$.

3.4 Evaluation and mechanism stress test

The empirical audit uses $q \in \{0.05, 0.10\}$. Point estimates are accompanied by 500 paired full-size empirical resamples per outcome and q . Each resample uses common record multiplicities across all four cells while holding the observed scores and absolute budgets fixed. The resulting central 95% ranges measure conditional stability of the reported four-cell contrasts.

The controlled simulation crosses availability fractions 0.5, 0.7, and 0.9; risk–availability relationships in which higher risk is less available, independent, or more available; score noise levels 0.5, 1,

and 2; and $q \in \{0.05, 0.10, 0.20\}$. These factors yield 81 conditions. Each of 100 repetitions contains 100,000 evaluation records and an independent 50,000-record threshold-selection population. Outcomes, availability, and scores are shared across the compared policies within a repetition.

4 Results

4.1 Candidate eligibility and event ceilings

Early entry contains about 74.7% of target-population records, but only 71.8% of preterm events, 70.6% of NICU events, and 69.4% of low-birth-weight events. Before ranking begins, the early gate therefore leaves 28.2–30.6% of events outside its reachable population. Including every documented prenatal-care start in months 1–10 raises the event ceiling to 91.1–92.1%.

In absolute terms, early eligibility excludes 74,624–89,111 outcome events across the three tasks. Even the documented-start gate leaves 20,558–27,087 events outside the candidate population. These are reach constraints: changing the ranking leaves events outside its decision set unreachable.

The excluded records carry concentrated risk. Outcome prevalence among no-care records is 2.4–2.7 times the early-entry prevalence across the three targets; unknown-timing records are 1.7–1.9 times higher. Candidate expansion therefore changes both population reach and case mix, making a full-population event denominator essential.

Table 1: The candidate gate fixes reach before ranking. Early entry denotes prenatal-care initiation in months 1–3; documented start includes months 1–10. Risk ratios compare outcome prevalence in excluded groups with the early-entry group.

Outcome	Early eligible n	Early share	Early ceiling	Documented ceiling	No-care / unknown risk
Preterm birth	2,601,060	74.7%	71.8%	91.1%	$2.7 \times / 1.8 \times$
NICU admission	2,597,274	74.7%	70.6%	92.1%	$2.4 \times / 1.7 \times$
Low birth weight	2,600,093	74.7%	69.4%	91.6%	$2.6 \times / 1.8 \times$

4.2 Eligibility–capacity accounting under top- q selection

Figure 1 and Table 2 connect the nominal operating point to its actual workload. Across outcomes, the selected count increases from approximately 260,000 to 348,000. Population event capture rises from 14.63% to 20.30% for preterm birth, 13.93% to 19.52% for NICU, and 16.06% to 22.83% for low birth weight. These are relative gains of 38.8%, 40.1%, and 42.2%, respectively.

The proportional policy adds 87,931–88,023 service slots per outcome and captures 16,505–17,177 additional events. Its marginal yield is therefore 18.8–19.5 additional events per 100 added slots, equivalent to 5.1–5.3 additional selections per extra captured event. Thus, the higher capture is obtained under an explicitly 33.8% larger service capacity.

Table 2: Same nominal top 10%, different operational result. Population capture uses the full target-population event denominator. ϕ_E and ϕ_B are the symmetric eligibility and capacity allocations; conditional empirical-resampling ranges appear in Appendix 7.

Outcome	Selected n	Workload	Capture (%)	Rel. gain	ϕ_E / ϕ_B (pp)	s_B
Preterm birth	260,106→348,038	+33.8%	14.63→20.30%	+38.8%	2.15/ 3.52	62.1%
NICU admission	259,727→347,658	+33.9%	13.93→19.52%	+40.1%	2.11/ 3.47	62.2%
Low birth weight	260,009→348,032	+33.9%	16.06→22.83%	+42.2%	2.60/ 4.17	61.6%

The four-cell audit changes the interpretation of those gains. Symmetric accounting assigns 2.11–2.60 percentage points to eligibility and 3.47–4.17 points to additional capacity, placing 61.6–62.2% of the joint contrast on the workload increase. Holding the early budget fixed yields eligibility gains of 1.89–2.27 points; holding the all-record budget fixed yields 2.32–2.93 points. Both are materially smaller than the joint 5.59–6.77-point cohort-relative contrast.

The pattern persists at $q = 5\%$, where capacity receives 63.9–65.6% of the symmetric allocation. Conditional ranges remain above 60.4% for every target and operating point. A random ranking allocates 79.7–87.9% to capacity at $q = 10\%$, showing that a majority capacity allocation can arise mechanically under a proportional budget. The learned ranking shifts more of the joint gain toward eligibility because the added candidate groups contain greater event mass. Complete four-cell counts and both operating points appear in Appendix B.

4.3 Fixed-budget and transported-threshold policies

Table 3 compares three policies on the same frozen score vector. Cohort-relative top-10% evaluation jointly changes eligibility and workload. Fixing the all-record absolute budget isolates candidate expansion: capture increases by 2.32–2.93 points, with exactly the same number selected on both sides. A threshold chosen once on the CDC 2022 validation split selects 9.15–9.82% of early-entry records and 10.20–10.40% of all records in CDC 2023. Under that transported threshold, the all-record workload is 1.42–1.49 times larger and capture increases by 6.36–8.19 points.

Table 3: One score vector, three resource policies at the nominal 10% operating point. Counts and population capture compare early-entry with all-record eligibility. The score threshold is chosen on CDC 2022 and transported unchanged to CDC 2023.

Outcome	Resource policy	Selected n : early $\rightarrow$ all	Capture: early $\rightarrow$ all	Δ (pp)
Preterm birth	Cohort-relative top 10%	260,106 $\rightarrow$ 348,038	14.63 $\rightarrow$ 20.30%	+5.67
	Fixed absolute b_a	348,038 $\rightarrow$ 348,038	17.89 $\rightarrow$ 20.30%	+2.40
	Transported 2022 threshold	245,487 $\rightarrow$ 361,709	14.06 $\rightarrow$ 20.84%	+6.78
NICU admission	Cohort-relative top 10%	259,727 $\rightarrow$ 347,658	13.93 $\rightarrow$ 19.52%	+5.59
	Fixed absolute b_a	347,658 $\rightarrow$ 347,658	17.20 $\rightarrow$ 19.52%	+2.32
	Transported 2022 threshold	255,166 $\rightarrow$ 361,714	13.76 $\rightarrow$ 20.12%	+6.36
Low birth weight	Cohort-relative top 10%	260,009 $\rightarrow$ 348,032	16.06 $\rightarrow$ 22.83%	+6.77
	Fixed absolute b_a	348,032 $\rightarrow$ 348,032	19.90 $\rightarrow$ 22.83%	+2.93
	Transported 2022 threshold	237,973 $\rightarrow$ 355,034	14.98 $\rightarrow$ 23.17%	+8.19

These contrasts differ because they implement different resource commitments on a shared ranking. A complete cross-cohort claim therefore states its candidate rule, selected count, and event denominator alongside the gain.

The cross-year workload check points in the same direction. Transporting the stored CDC 2022 top-10% score cutoffs to target-specific CDC 2024 cohorts selected 10.65% of preterm records, 9.50% of NICU records, and 8.30% of low-birth-weight records. Relative to a 10% service budget, these rates create 22,668 additional preterm slots but leave 17,621 NICU and 59,716 low-birth-weight slots unfilled. A fixed score cutoff therefore defines a workload that can move across years; the exact selected count remains the operational quantity for capacity planning.

4.4 Controlled mechanism stress test

Figure 3 maps the protocol difference across all 81 synthetic conditions. Eighteen conditions report a positive cohort-relative gain while candidate expansion at the all-record absolute budget reduces population capture. Nine have a central replicate range entirely below zero. Every reversal arises when higher-risk records are preferentially available, so the newly admitted pool is lower risk and can displace selected events.

At 70% availability, medium score quality, and $q = 10\%$, for example, cohort-relative capture increases 5.52 points while fixed-budget candidate expansion changes capture by -0.28 points; 93 of 100 repetitions reverse. The largest mean absolute interaction is 9.49 points. In CDC 2023, all six fixed-budget eligibility contrasts remain positive, so the ambiguity changes effect magnitude. The controlled grid shows that selective availability can also change effect direction.

4.5 Temporal score diagnostics

The four-cell accounting fixes one CDC 2023 score vector, isolating its within-year policy contrasts from score drift. A temporal check additionally shows that the frozen 2022 pipelines retain similar

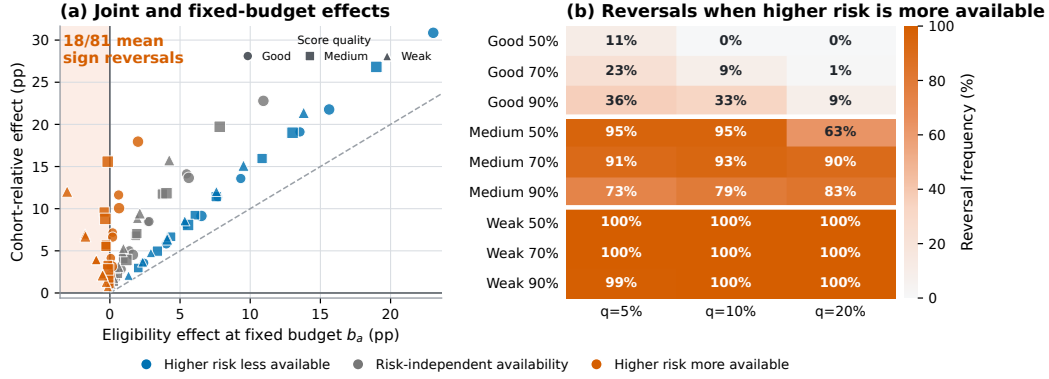


Figure 3: Full-grid mechanism stress test. Panel (a) compares the cohort-relative conclusion with candidate expansion at the all-record-derived absolute budget b_a . The shaded quadrant contains sign reversals: the joint top- q contrast is positive while fixed-budget eligibility expansion is negative. Panel (b) gives replicate-level reversal frequency for conditions where higher risk is more available. Marker shape denotes score quality.

aggregate discrimination in CDC 2023 and provide a transported ranking for the policy audit. Across outcomes, ROC-AUC changes by at most 0.003 and average precision by at most 0.003 from the CDC 2022 validation split to CDC 2023. CDC 2023 Brier skill relative to a prevalence-only forecast is 1.7–2.4%. These diagnostics anchor the empirical audit in transported scores while preserving the decision policy as the primary estimand.

Table 4: Frozen-score temporal check. Brier baseline is the constant CDC 2023 prevalence prediction; skill is $1 - \text{Brier}_{\text{model}} / \text{Brier}_{\text{baseline}}$.

Outcome	ROC-AUC: 2022 $\rightarrow$ 2023	AP: 2022 $\rightarrow$ 2023	Brier: model / baseline	Brier skill
Preterm birth	0.614 $\rightarrow$ 0.612	0.146 $\rightarrow$ 0.146	0.0780 / 0.0795	1.9%
NICU admission	0.611 $\rightarrow$ 0.608	0.139 $\rightarrow$ 0.141	0.0783 / 0.0796	1.7%
Low birth weight	0.653 $\rightarrow$ 0.652	0.133 $\rightarrow$ 0.132	0.0636 / 0.0651	2.4%

4.6 Sensitivity and robustness

Two feature ablations preserve the allocation pattern. Removing maternal race and refitting each outcome model at $q = 10\%$ leaves capacity shares of 60.9–62.4%. A separate eight-feature specification removes prior other pregnancy outcomes (prior_terminations) and repeats the complete audit at both budgets with paired resampling. Capacity still accounts for 61.5–64.3% of the joint contrast, and all interaction ranges remain above zero (+0.32 to +0.69 percentage points). The result therefore persists across two alternative conservative landmark specifications.

5 Discussion

Resource interpretation of top- q evaluation. Top- q evaluation is operationally coherent when service volume scales with the candidate population; a fixed absolute budget answers the constrained capacity question. The distinction is material here: the headline 38.8–42.2% relative capture gain includes a 33.8% workload increase. The four-cell design makes that resource policy visible and supplies both fixed-budget contrasts.

Separating reach, ranking, and capacity. The candidate gate determines the event ceiling, ranking determines which eligible cases approach the cutoff, and capacity determines how many positions can be filled. Freezing the score vector isolates broader reach at fixed capacity from proportional resource expansion; the interaction records when the value of one lever depends on the other. Transported score cutoffs form a third regime in which workload emerges from the subsequent score distribution, as the CDC 2024 check illustrates.

Applicability across deterministic score vectors. The four-cell accounting accepts any fixed deterministic score vector, and capacity monotonicity follows from the nesting of top- b selections independently of the LightGBM functional form. Candidate expansion at fixed b depends on which scores and events enter the pool, so its sign remains empirical. The random-ranking reference and the controlled grid spanning three score-noise levels recover both properties: capacity usually explains most of a proportional-budget contrast, while selective availability can reverse the fixed-budget effect. Model choice changes where a study lies in this mechanism map; the audit defines the decision comparison around it.

A recommended reporting profile. Based on the empirical and controlled audits, Table 5 proposes a minimum reporting checklist for cross-cohort top- q comparisons. The checklist links each reported number to the target population, candidate gate, resource rule, or decomposition needed to interpret the operating point.

Table 5: Minimum reporting checklist for cross-cohort top- q evaluation. The final column illustrates each item with the CDC 2023 audit.

Required item	Minimum report	Example from this study
Target population	Population rule, N , total events E , and event denominator	CDC 2023 singleton births; $N \approx 3.48\text{M}$; all target events remain in the denominator
Candidate gate	Eligibility rule, eligible n , E_S , and event ceiling E_S/E	Care entry in months 1–3; $n \approx 2.60\text{M}$; ceiling 69.4–71.8%
Resource rule	Fraction q or absolute budget b , with exact selected n	Top 10%; selected n : 260k→348k (+33.8% workload)
Operating result	Precision, eligible recall, and population event capture	Capture: 13.93–16.06%→19.52–22.83%
Cross-cohort audit	Four cells, fixed-budget contrasts, interaction, and symmetric allocation	Capacity receives 61.6–62.2% of the joint allocation

The profile applies wherever a score becomes a finite worklist: imaging queues, screening, case management, and public-sector allocation all require a declared fixed, proportional, or threshold-driven capacity rule. Exact selected counts prevent a stable percentage from disguising a changing workload.

Implementation in existing evaluation pipelines. Stored identifiers, outcomes, candidate masks, and one score vector are enough to construct the four cells. Implementations record both budgets, selected events, and the full-population denominator, then reuse common record weights for paired resampling. Aggregate counts and configurations provide an auditable path from a top- q result to its service policy without releasing row-level clinical data.

6 Conclusion

Top-fraction comparisons across cohorts encode service decisions in addition to ranking summaries. In the CDC audit, the same nominal top 10% expands the workload by one third, and symmetric accounting assigns about 62% of the capture gain to that capacity increase. Controlled experiments identify conditions in which cohort-relative and fixed-budget conclusions have opposite signs. Reporting the target population, candidate rule, exact selected count, full-population event denominator, and fixed-budget contrast makes the policy claim auditable. The present scope is a birth-certificate score and a care-entry eligibility proxy; prospective EHR landmarks, subgroup budgets, and additional temporal cohorts provide the next transport tests.

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325 A Outcome, Model, and Cohort Details

Table 6: Baseline-oriented birth-certificate variables used by the frozen ranking instrument. Encounter-specific availability at months 1–3 requires validation in a prospective EHR source.

Feature	CDC source meaning
Maternal race	Maternal characteristic on birth record
Maternal height	Maternal characteristic on birth record
Pre-pregnancy BMI	Derived from pre-pregnancy measures
Pre-pregnancy weight	Weight before pregnancy
Previous live births now living	Previous live-born infants still living
Previous live births now dead	Previous live-born infants no longer living
Other pregnancy outcomes	Non-live-birth outcomes; may include current pregnancy
Pre-pregnancy diabetes	Diabetes diagnosed before this pregnancy
Pre-pregnancy hypertension	Hypertension diagnosed before this pregnancy

326 For CDC 2023 preterm records, early entry contains 2,601,060 records and 217,632 events; later
327 entry contains 739,298 and 58,282; no care contains 76,520 and 17,387; unknown timing contains
328 63,504 and 9,700. Target-specific counts differ slightly because each outcome has its own missing-
329 label exclusions. Source code 99 is normalized to missing before cohort construction, while source
330 value 0 denotes no prenatal care.

331 B Complete Four-Cell Results

332 The point-estimate budgets at $q = 5\%$ are 130,053/174,019 for preterm, 129,864/173,829 for NICU,
333 and 130,005/174,016 for low birth weight (early/all-record). At $q = 10\%$ they are 260,106/348,038,
334 259,727/347,658, and 260,009/348,032. These observed integer budgets remain fixed across empir-
335 ical resamples.

336 C Synthetic Grid Details

337 The synthetic experiment crosses three availability fractions, three risk–availability slopes, three
338 score-noise levels, and three selection fractions. Intercepts are solved numerically so the target

Table 7: Complete symmetric-allocation results. All effects are percentage-point changes in population event capture; brackets give central 95% paired empirical-resampling ranges conditional on frozen scores and observed budgets.

Outcome	q	Joint Δ	ϕ_E	ϕ_B	Symmetric share s_B	Interaction
Preterm birth	5%	3.37 [3.30, 3.43]	1.16 [1.09, 1.22]	2.21 [2.17, 2.25]	65.6 [64.3, 67.1]%	0.33 [0.26, 0.41]
Preterm birth	10%	5.67 [5.58, 5.77]	2.15 [2.06, 2.24]	3.52 [3.48, 3.57]	62.1 [61.1, 63.3]%	0.51 [0.44, 0.60]
NICU admission	5%	3.31 [3.26, 3.38]	1.19 [1.13, 1.25]	2.12 [2.09, 2.16]	64.1 [62.8, 65.4]%	0.29 [0.23, 0.36]
NICU admission	10%	5.59 [5.51, 5.67]	2.11 [2.03, 2.20]	3.47 [3.44, 3.52]	62.2 [61.1, 63.3]%	0.42 [0.34, 0.51]
Low birth weight	5%	3.96 [3.89, 4.06]	1.43 [1.34, 1.52]	2.53 [2.50, 2.57]	63.9 [62.3, 65.4]%	0.40 [0.32, 0.49]
Low birth weight	10%	6.77 [6.67, 6.89]	2.60 [2.49, 2.72]	4.17 [4.12, 4.22]	61.6 [60.4, 62.7]%	0.66 [0.55, 0.78]

Table 8: Auditable four-cell point estimates. Each cell reports population event capture in percent followed by selected true events in parentheses. The two budgets are exact selected counts.

Outcome	q	Budgets b_e/b_a	$C_{00}: S_e, b_e$	$C_{01}: S_e, b_a$	$C_{10}: S_a, b_e$	$C_{11}: S_a, b_a$
Preterm birth	5%	130,053/174,019	9.05 (27,434)	11.10 (33,618)	10.04 (30,432)	12.42 (37,631)
Preterm birth	10%	260,106/348,038	14.63 (44,326)	17.89 (54,216)	16.52 (50,061)	20.30 (61,503)
NICU admission	5%	129,864/173,829	8.51 (25,801)	10.48 (31,795)	9.55 (28,967)	11.82 (35,854)
NICU admission	10%	259,727/347,658	13.93 (42,252)	17.20 (52,160)	15.84 (48,033)	19.52 (59,201)
Low birth weight	5%	130,005/174,016	9.53 (23,229)	11.86 (28,918)	10.76 (26,224)	13.49 (32,892)
Low birth weight	10%	260,009/348,032	16.06 (39,145)	19.90 (48,505)	18.33 (44,679)	22.83 (55,650)

339 marginal prevalence or availability fraction holds before Monte Carlo variation. Common random
340 numbers are used across all 81 conditions within each repetition, and the available-subset and all-
341 population policies share outcomes and score vectors. Reported 2.5th and 97.5th percentiles describe
342 the replicate distribution.

343 The score-noise levels $\sigma_R = 0.5, 1, 2$ imply expected correlations between score and latent risk of
344 approximately 0.894, 0.707, and 0.447. The largest absolute replicate-level interaction is 10.72 per-
345 centage points. Full condition summaries and replicate outputs are retained in the prepared release
346 bundle.

347 D Audit Identities and Artifacts

348 **Deterministic identity checks.** For fixed S , scores, and deterministic tie breaking, $\text{Top}(S, b_e) \subseteq$
349 $\text{Top}(S, b_a)$ whenever $b_e \leq b_a$. Increasing the budget adds exactly $b_a - b_e$ ranked positions and
350 between zero and $b_a - b_e$ events. Both marginal capacity contrasts are therefore non-negative, as
351 is their average ϕ_B . Substituting Equation 6 also gives $\phi_E + \phi_B = C_{11} - C_{00}$ exactly. These
352 statements concern a shared score vector; retraining introduces an additional factor.

353 **Artifacts.** All manuscript figures and tables are generated from included CSV files. Verification
354 scripts assert the six primary Shapley rows, 500 paired empirical resamples per target and budget, ex-
355 act accounting identities, all 81 synthetic conditions, 18 mean sign reversals, and 9 conditions whose
356 central replicate range is negative. The prepared release bundle contains aggregate inputs, figure and
357 table generators, verification scripts, the locked synthetic configuration and runner, synthetic repli-
358 cate outputs, and the temporal threshold-transport script. It redistributes no individual-level Natality
359 records. Use of the public-use files follows the NCHS Data User Agreement [National Center for
360 Health Statistics, 2025]; LightGBM is distributed under the MIT License [Microsoft Corporation
361 and LightGBM Contributors, 2026].

362 **Raw-data reconstruction.** The CDC 2022 split is drawn before target-specific exclusions with
363 `default_rng(2026)`: 20% enters validation and the remainder enters training, on which prepro-
364 cessing is fitted. Unknown outcomes are excluded target-wise, and original row order breaks score
365 ties. Thresholds are selected on the target-specific CDC 2022 all-record validation pool and applied
366 with $\geq$. Empirical and synthetic seeds are 2027 and 2028. Section 3 and Appendices A–C specify
367 the remaining cohort, model, and estimand details.

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